Natural History

Alveary

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Science

Natural History: Grade 8

GRADE(S): 8

About the Course

Learners explore considerations from the history of microbiology and infectious disease. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 8

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context, and with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 8
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

Natural History: Grade 8
For approximately eighth-grade students based on a balance of complexity and reading level. However, learners may be freely combined based on needs and preferences.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
8	Natural History: Grade 8 1 time/week 30 min	Invincible Microbe

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 8				

Natural History: Grade 8	General Science: Grade 8 Nature Walks & Scouting: Grades 1-8	General Science: Grade 8	General Science: Grade 8 Nature Notebook: Grade 8	Labs: Grade 8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 8
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 8
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

Natural History: Grade 8
☐ Any term: a science museum with an exhibit on microbes

Term Prep & Teacher Tips

Natural History: Grade 8
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• 8 Tbsp cornstarch
• 1-2 c milk (Term 1)
• 3-5 c white vinegar
• paper towels
• water
• 1 egg (Term 1)
• 3 antimicrobial agents used at home (e.g., soap, hydrogen peroxide, rubbing alcohol, essential oils, etc.)

Reminders

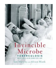
Natural History: Grade 8
☐ Note that the first lesson of Natural History is an activity, so be prepared with these materials and any others from the supply lists on day 1.
☐ Any type of milk is needed for Week 5. Make a note on your calendar if you will need to purchase this closer to the appropriate time.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

- [View Book List Details]

Natural History: Grade 8



Invincible Microbe

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- [View Basic Supplies]
- [View Supply List Details]

Natural History: Grade 8

Prepared Microscope Slides

Household Items - Natural History: Grade 8

Slides and Coverslips

Student Microscope

Quick Links

Click THIS text or scan the QR code for links.

Related Course Quick Links

- Extra Helpings
- Outdoor Work
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android
- Alternate One Year Physical Science Lesson Plans
- Foundations (See Section 13: Science)

Natural History: Grade 8

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

Natural History: Grade 8

Click THIS text or scan the QR code for links.

Term 1

WEEK 1

⬚ **30m Natural History: Grade 8 - Lesson 1**
Cell Activity

▯ Materials: toothpick, glass slide and cover slip, candle, methylene blue stain, water, paper towel, microscope, egg, cup/glass, white vinegar

→ INTRO
Most of your experience in natural history has been with Things that you can see around you. The invention of the microscope led to the discovery of a whole new world. This world of microscopic life is our focus this year. We begin by better understanding the microscopic cells from which you are made. First, you will make a slide of your cheek cells. Then you will set up an egg as a simple model of the cell.

→ ACTIVITY
• Using a toothpick, gently rub the inside of your cheek.

• Now rub the toothpick on the surface of a glass slide, keeping the smear near the middle of the slide. Let this dry for a moment on a paper towel while you light your candle.

• Heat fix your slide by passing the slide through the flame 3-4 times. The underside of the slide should pass through the flame, but never stop. This will help your sample stick to the slide better. When you are finished, you can blow out the candle.

• Using protection, squeeze a drop of methylene blue stain onto your cheek smear. This stain will stain everything, so you MUST protect your clothing and anything else you do not want to be stained! Let the slide with the drop of stain sit for about 2 minutes.

• Gently rinse from the back of the slide with a little clean water and carefully blot dry. Do not rub your smear.

→ OBSERVE & NARRATE
• Set a cover slip on your slide and examine it under the microscope, slowly increasing the power as you want to see more. Draw or diagram what you see in your science notebook.

→ ACTIVITY
• Now, place an egg inside a cup or glass and cover with vinegar. Replace the vinegar every day for three days. Once the shell has been dissolved, the egg can be left on a paper towel until your next lesson.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ **30m Natural History: Grade 8 - Lesson 2**
Cell Activity

▯ Materials: egg from last week, Prepared Microscope Slide Set, microscope, image link

PREP: Read Teacher Tip

→ INTRO
Today, you will compare your egg model to the cheek cells that you observed under the microscope, compare these to a variety of animal cells, and then to some bacterial cells.

→ OBSERVE, NARRATE, & DISCUSS
• Notice that your egg model has an outer membrane and a well-formed internal structure (the yolk) that is suspended in a matrix (the egg white). Compare to your cheek cells from last week. How are they similar? How

★ TEACHER TIP
If needed, be sure to pour your agar plates for next week according to the package instructions.
∞ Video Link: Pouring Agar Plates (2:51)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Term 1

are they different?

• This same basic structure is used to accomplish a great number of tasks. View slides 14-20 from your Prepared Microscope Slide Set. Try to identify these basic parts in each slide. Look carefully at slide 15. Red blood cells lose their nucleus so that they can carry oxygen more efficiently. You may also be able to find some white blood cells on this slide. How do they look different?

• View the Cellular Diversity Slides at the link to see a variety of bacterial cells. What do you notice about the bacterial cells in comparison? Can you still identify the same basic parts? How do the cells compare in size? Draw or diagram your observations in your science notebook
∞ Link: Cellular Diversity Slides

WEEK 3

⬚ **30m Natural History: Grade 8 - Lesson 3**
Microbe Activity

▯ Materials: agar plates

→ INTRO
Because bacteria are so small, humans were unaware of their existence or their place in our world for a very long time. One way to see them more easily is to grow a colony of them. You will use petri dishes with a kind of food, called agar, to do this. The dishes will be left open at various locations for several hours to see what kinds of microorganisms are in the environment at each location.

→ ACTIVITY
• Choose three locations that you would like to test for the presence of bacteria. These might be your classroom, the bathroom, the kitchen, the playground, etc.

• Take four of your agar plates and, keeping them closed, turn them upside down on your work surface. Label the bottom of each plate using a permanent marker. One plate will be labeled "control," while the remaining three will be labeled with your three locations.

• Leave the control plate closed and where it is while taking the three experimental plates to their locations. For each experimental plate, open the lid and leave the plate right-side-up in that location. You will leave it there for about three hours.

• After several hours exposed, collect each plate, replacing the lid. You may want to secure all of the plates with tape. Then leave them in a warm (but not hot) place for the next week.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.